THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If $P(E) = 0.35$ and $P(F) = 0.45$ and E, F cannot occur together, then $P(E \text{ or } F) =$ 1

(a) 0.20 **(b)** 0.1575 **(c)** 0.80 **(d)** 0.10

2. The nth term of an AP is $t_n = 3n + 2$. The sum of its first 16 terms is: 1

(a) 437 **(b)** 456 **(c)** 443 **(d)** 440

3. Parallel chords of lengths 30 cm and 40 cm lie on opposite sides of the centre of a circle of radius 25 cm. The distance between them (in cm) is: 1

(a) 37 **(b)** 34 **(c)** 36 **(d)** 35

4. The area of a cyclic quadrilateral with sides 5 cm, 8 cm, 19 cm and 20 cm is (cm^2): 1

(a) 128 **(b)** 125 **(c)** 127 **(d)** 126

5. Two dice are rolled together. The probability that the product of the numbers is 6 is: 1

(a) $\frac{1}{10}$ **(b)** $\frac{2}{9}$ **(c)** $\frac{1}{9}$ **(d)** $\frac{8}{9}$

6. The 5th term of an AP is 44 and the 12th term is 16. The 17th term is: 1

(a) -12 **(b)** 0 **(c)** -4 **(d)** -8

7. Parallel chords of lengths 30 cm and 40 cm lie on the same side of the centre of a circle of radius 25 cm. The distance between them (in cm) is: 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 4

8. A circle of radius 14 cm (take $\pi = \frac{22}{7}$) is inscribed in a square. The area of the square not covered by the circle is (cm^2): 1

(a) 169 **(b)** 167 **(c)** 170 **(d)** 168

9. Two dice are rolled together. The probability that the sum is a prime number is: 1

(a) $\frac{5}{12}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{5}{13}$ **(d)** $\frac{7}{12}$

10. The sum of the first n terms of a sequence is $S_n = 3n^2 + 1n$. Its 12th term is: 1

(a) 67 **(b)** 73 **(c)** 76 **(d)** 70

11. Two circles of radii 13 cm and 20 cm intersect in a common chord of length 24 cm and their centres lie on opposite sides of it. The distance between the centres (in cm) is: 1

(a) 23 **(b)** 22 **(c)** 21 **(d)** 20

12. Two concentric circles have radii 21 cm and 35 cm (take $\pi = \frac{22}{7}$). The area of the ring between them is (cm^2): 1

(a) 2463 **(b)** 2464 **(c)** 2465 **(d)** 2466

13. **Assertion (A):** The probability of an impossible event is 0. 1
Reason (R): The probability of any event lies between 0 and 1, both included.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** The sequence $2, 4, 8, 16, \ldots$ is a geometric progression. 1
Reason (R): The sequence $1, 3, 5, 7, \ldots$ is an arithmetic progression.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The opposite angles of a cyclic quadrilateral are supplementary. 1
Reason (R): The opposite angles of a cyclic quadrilateral are equal.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Three coins are tossed. Find the probability of getting at least one head, using the complement rule. 2

17. If $S_n = 3n^2 + 4n$ is the sum of the first n terms of a sequence, find its first term and its 10th term. 2

18. Two parallel chords of lengths 30 cm and 40 cm lie on the opposite sides of the centre of a circle of radius 25 cm. Find the distance between them. 2

19. Find the area of a cyclic quadrilateral with sides 5, 8, 19 and 20 cm. 2

20. Two dice are rolled together. Find the probability that the product of the numbers is 6. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A bag contains red, blue and green balls. $P(\text{red}) = \frac{1}{5}$ and $P(\text{blue}) = \frac{2}{5}$ and there are 40 balls in all. Find $P(\text{green})$ and the number of balls of each colour. 3

22. The sum of the 2nd and 10th terms of an AP is 64 and its 5th term is 28. Find the AP's first term and the sum of its first 14 terms. 3

23. Parallel chords of lengths 16 cm and 30 cm in a circle of radius 17 cm. Find the distance between them in both possible positions. 3

24. A running track is made of a rectangle 80 m $\times$ 14 m with a semicircle (radius 7 m) on each of the two shorter sides (take $\pi = \frac{22}{7}$). Find the length of the track along its boundary and the area enclosed. 3

OR

In 200 throws of a die, the number of times a prime number appeared was 108 and a number greater than 4 appeared 63 times (these include 30 throws showing 5). Find the experimental probability of 3

(i) a prime number,

(ii) a number greater than 4,

(iii) a prime number or a number greater than 4.

25. Find the sum of all natural numbers between 20 and 300 that leave remainder 3 when divided by 7. 3

OR

AB is a diameter of a circle; C, D lie on the circle on the same side of AB so that $ABCD$ is a cyclic quadrilateral with $\angle DAB = 41^\circ$. Find $\angle BCD$, $\angle ADB$ and $\angle ABD$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** In a Ludo game at home, Arjun rolls a fair six-faced die once. 5

(i) How many outcomes are there in the sample space?

(ii) Find the probability of getting an even number.

(iii) Find the probability of getting a number greater than 4.

(iv) Find the probability of getting a multiple of 3 and hence the probability of not getting a multiple of 3.

27. Two dice are rolled together. Find the probability that 5

(i) the sum is 7,

(ii) the sum is more than 7,

(iii) the sum is at most 7. Verify that

(ii) and

(iii) add to 1.

OR

A contractor pays a penalty for delay: ₹100 on day 1, ₹125 on day 2, ₹150 on day 3 and so on. If the work is 24 days late, find the penalty on the last day and the total penalty. 5

END OF QUESTION PAPER — Total: 50 Marks

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